A Consensus-Optimization Framework for Progressive Clearance-Aware PCB Legalization with Irregular Components

ABSTRACT

In modern printed circuit board (PCB) designs, legalization must resolve overlaps among heterogeneous, irregularly shaped, and arbitrarily oriented components on double-sided boards while minimizing design rule violations (DRVs) and preserving placement quality. Existing constraint graph-based methods rely on mixed-integer quadratic programming (MIQP), which treats geometric feasibility, clearance satisfaction, and displacement minimization as a single tightly coupled problem; this suffers from scalability limitations and, when no globally feasible solution is found within the runtime budget, often leaves the placement essentially unchanged in high-density designs. This paper proposes a consensus-optimization framework for progressive clearance-aware PCB legalization that decomposes the tightly coupled problem into coordinated subproblems—displacement minimization, geometric feasibility, and pad-level clearance enforcement—and drives them toward a globally consistent legal placement. The framework is realized through an Alternating Direction Method of Multipliers (ADMM) scheme, in which a quadratic program minimizes displacement while a SAT-based minimum translation vector (MTV) subproblem resolves collisions and enforces net-aware, per-pad clearance directly on exact irregular geometries. Experimental results on industrial PCB benchmarks demonstrate that the proposed framework reduces geometric violations by 55.72% in overlap and 21.61% in clearance on average, significantly outperforming MIQP-based approaches and continuing to improve highly congested cases where the discrete baseline stagnates. Although this comes at the cost of larger displacement and a moderate wirelength increase, the framework preserves placement quality and achieves lower average runtime.

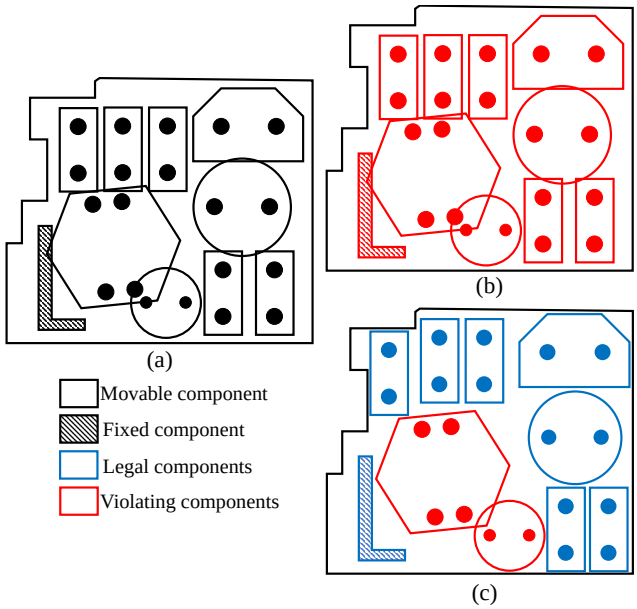


Figure 1: Comparison of the baseline MIQP-based legalizer and the proposed consensus-optimization framework for PCB legalization. (a) Initial placement from global placement with severe overlaps and rule violations. (b) MIQP-based legalization stagnates due to combinatorial complexity and fails to produce meaningful updates within practical runtime. (c) The proposed consensus-optimization legalization continuously reduces DRVs and progressively approaches feasibility.

1 INTRODUCTION

With the increasing complexity of modern electronic systems, printed circuit board (PCB) designs have become significantly more challenging due to high component density, heterogeneous integration, and irregular board geometries. Compared to integrated circuit (IC) placement, PCB placement must explicitly handle irregular-shaped components, arbitrary orientations, complex clearance rules, and non-convex board boundaries. These requirements make *legalization*—the process of removing overlaps while minimizing displacement—a critical yet computationally expensive task, since the tight coupling between geometric feasibility and electrical constraints means that resolving one violation may easily introduce new conflicts elsewhere. In practice, components may mix surface-mount pads and through-hole structures, so their interactions depend not only on geometry but also on net relationships and layer assignments. Moreover, modern designs frequently operate at high utilization, where even small displacements may introduce new violations.

These characteristics imply that an effective PCB legalizer must be both geometry-aware and globally coordinated: purely local heuristics fail to resolve tightly coupled conflicts, while purely global formulations become computationally intractable. Although global placement techniques exist for PCB design, their solutions often miss the fine-grained geometric and rule-driven considerations needed in detailed refinement, so the initial placement may still contain substantial overlaps and rule violations, requiring a robust legalization stage.

Existing PCB legalization approaches primarily rely on simplified geometric representations or discrete optimization frameworks. Rectangular-based approximations introduce geometric inaccuracies when handling irregular components, potentially leading to pseudo-overlaps or sub-optimal area utilization. Conversely, discrete formulations, such as constraint graphs, mixed-integer programming (MIP), or mixed-integer quadratic programming (MIQP), restrict the solution space and often suffer from scalability limitations in modern high-density designs. These limitations become more pronounced when the initial placement is far from feasibility: the combination of tight geometric constraints and discrete decision variables may render the problem extremely difficult to solve within practical runtime limits, so the solver may terminate early without producing a meaningful placement update.

This motivates a legalization framework that can simultaneously handle exact geometries and large-scale optimization. As illustrated in Fig. 1, the initial global placement (GP) contains severe overlaps and rule violations (a); the MIQP-based legalizer fails to find a feasible solution within practical runtime limits, yielding little or no improvement (b); in contrast, the proposed consensus-optimization method iteratively reduces DRVs and progressively improves the placement even when full feasibility is not immediately achieved (c).

PCB legalization faces several unique challenges compared to IC placement. As illustrated in Fig. 2, components may have irregular shapes and arbitrary orientations, leading to complex geometric interactions. In addition to non-overlap constraints, PCB designs require strict clearance enforcement between components, pads, and nets, and all components must be placed within non-convex board boundaries. Unlike traditional monolithic formulations, it is therefore desirable to decompose legalization into two complementary aspects: displacement minimization, which is typically smooth and convex, and geometric feasibility enforcement, which involves highly nonconvex constraints induced by irregular geometries and clearance requirements. A framework that can effectively separate and coordinate these two aspects is essential for achieving both scalability and robustness in modern PCB legalization.

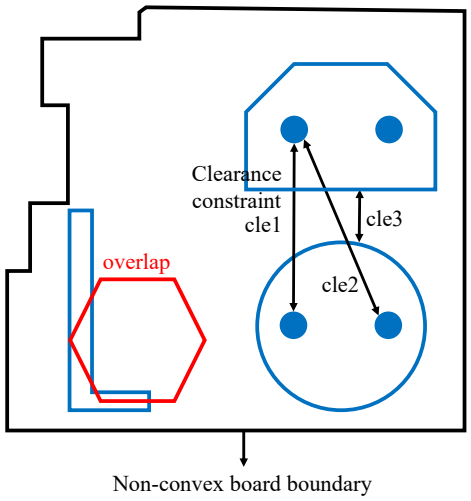


Figure 2: Key challenges in PCB legalization, illustrated on a single layout. Irregular-shaped components may overlap in complex ways that cannot be accurately captured by rectangular approximations; all components must remain within the possibly non-convex board boundary, which further restricts feasible placement regions; and clearance constraints require maintaining minimum spacing between different pads (cle1 and cle2) and between components (cle3), introducing additional geometric constraints.

1.1 Previous Works

PCB legalization has been extensively studied from both geometric modeling and optimization perspectives. Existing approaches can be broadly categorized into three groups: (1) topology-based methods with simplified geometric representations, (2) geometry-aware methods that improve modeling accuracy, and (3) discrete optimization methods based on constraint formulations.

Topology-based methods. Early approaches largely inherit techniques from IC placement, where components are simplified as rectangles and non-overlap constraints are enforced through topological representations such as constraint graphs, transitive closure graphs (TCG), and corner stitching [1–3]. While these methods are computationally efficient, their reliance on rectangular abstractions limits their applicability to PCB designs with irregular geometries and complex design rules.

Geometry-aware methods. To improve geometric accuracy, several works extend traditional representations to better capture PCB-specific characteristics. B*-tree-based methods have been adapted to support irregular board outlines and component geometries [4, 5], and window-based techniques further improve local refinement by resolving overlaps within confined regions [6]. Some methods decompose irregular components into multiple rectangular blocks, while others employ exact geometric constructs such as no-fit polygons (NFP) [7, 8]. Although these approaches provide higher geometric fidelity, they introduce significant computational overhead and are typically designed for packing problems rather than iterative legalization frameworks.

Discrete optimization methods. Another line of research formulates legalization as an optimization problem with explicit constraints. Heuristic approaches such as Abacus and Tetris [9, 10] achieve high efficiency but lack flexibility in handling complex design rules. Analytical methods model legalization as a continuous optimization problem using techniques such as KKT conditions or augmented Lagrangian frameworks, and more recently ADMM-based methods have been introduced to improve scalability [11, 12]. Among these approaches, constraint graph-based methods formulate legalization as MIQP problems to explicitly model relative placement and design constraints. In particular, the recent work [13] proposes a constraint graph-based PCB legalization framework using MIQP to handle dense, heterogeneous, irregular-shaped, and arbitrarily oriented components. While this formulation provides high modeling flexibility, it suffers from scalability limitations due to the combinatorial nature of discrete variables, and may fail to effectively reduce DRVs in high-density designs.

Overall, none of the existing approaches simultaneously achieves scalability, exact geometric modeling, and robust feasibility handling in dense PCB designs. This gap motivates the proposed consensus-optimization framework.

1.2 Our Contributions

In this paper, we propose a novel consensus-optimization framework for clearance-aware PCB legalization, in which ADMM serves as the underlying solver rather than the central contribution. Our major contributions are summarized as follows:

- We reformulate PCB legalization—conventionally solved as a single tightly coupled optimization—as a consensus-optimization framework that decouples displacement minimization, geometric feasibility, and clearance satisfaction into independently solvable subproblems coordinated toward a globally consistent legal placement, improving scalability over discrete MIQP-based methods and remaining extensible to additional design-rule constraints.
- We introduce a clearance-aware, geometry-independent feasibility model in which the legal region is defined by clearance-expanded (offset) geometries rather than raw polygons, integrating overlap removal and net-aware, per-pad clearance into a unified SAT-based minimum translation vector (MTV) formulation that handles exact irregular, concave, and arbitrarily oriented components and non-convex boundaries without conservative bounding-box approximations.
- We design a three-phase optimization flow—geometric preseparation, ADMM-based consensus optimization, and symmetric pair-based refinement—together with rigid heatsink-binding constraints, which progressively reduces violations rather than requiring a globally feasible solution, improving robustness in ultra-dense designs where exact discrete methods stagnate.
- Experimental results on industrial PCB benchmarks demonstrate that the proposed framework significantly reduces geometric violations (overlap and clearance) compared with a representative MIQP-based legalizer, achieving improvements on highly congested cases where the discrete baseline fails to make progress, while maintaining competitive wirelength and lower average runtime.

The rest of this paper is organized as follows. Section 2 introduces the preliminaries and problem formulation. Section 3 presents the proposed consensus-optimization legalization framework. Section 4 reports the experimental results. Finally, Section 5 concludes this paper.

2 PRELIMINARIES

This section establishes the mathematical foundations of the proposed framework, including the ADMM optimization structure and the geometric modeling techniques used for collision detection and resolution.

2.1 Consensus Optimization via ADMM

The proposed framework builds on consensus optimization, for which the Alternating Direction Method of Multipliers (ADMM) provides an efficient solver. ADMM is designed to solve constrained optimization problems of the form:

$$\min_{\mathbf{x},\mathbf{z}} f(\mathbf{x}) + g(\mathbf{z}) \quad \text{s.t. } A\mathbf{x} + B\mathbf{z} = \mathbf{c}. \tag{1}$$

The augmented Lagrangian for (1) is defined as:

$$\mathcal{L}_\rho(\mathbf{x}, \mathbf{z}, \boldsymbol{\lambda}) = f(\mathbf{x}) + g(\mathbf{z}) + \boldsymbol{\lambda}^T \mathbf{r} + \frac{\rho}{2}\|\mathbf{r}\|_2^2, \tag{2}$$

where $\mathbf{r} = A\mathbf{x} + B\mathbf{z} - \mathbf{c}$ is the primal residual and $\rho > 0$ is the penalty parameter. ADMM iteratively updates $\mathbf{x}$, $\mathbf{z}$, and the dual variable $\boldsymbol{\lambda}$ to enforce consensus.

A key advantage of ADMM is its ability to decompose a complex optimization problem into subproblems with different structures. In PCB legalization, the displacement objective is smooth and convex, while the geometric feasibility constraints are highly nonconvex due to irregular

Table 1: Summary of notations used in this paper

Notation	Description
$C, \mathcal{F}$	sets of components and fixed components
c_i	the i-th component in C
$\mathbf{x}_i, \mathbf{z}_i, \mathbf{u}_i$	position, auxiliary variable, and dual variable of component c_i
$\mathbf{x}_i^{\text{init}}$	initial position of component c_i
$\mathcal{S}_i$	set of sub-shapes of component c_i
s_i^a	the a-th sub-shape of component c_i
B	PCB board region
ρ, ϵ, κ	ADMM penalty parameter, tolerance, and overshoot factor
MTV	minimum translation vector

shapes and clearance requirements. Intuitively, the **x**-update pulls components toward their initial positions to preserve placement quality, while the **z**-update pushes components apart to eliminate overlaps and satisfy geometric constraints, and the dual update balances these two competing objectives so that the solution gradually converges to a feasible and stable configuration. This decomposition makes ADMM particularly suitable for large-scale PCB legalization, where directly solving a monolithic formulation would be computationally prohibitive.

Since the two updates are solved independently, additional geometric rules can be incorporated into the **z**-update without affecting the convexity of the **x**-subproblem, and the iterative nature of ADMM continuously reduces violations without requiring strict feasibility at intermediate steps.

2.2 Terminologies and Notations

The notations used throughout this paper are summarized in Table 1.

Let $C = \{c_1, \ldots, c_n\}$ denote the set of components. Each component c_i is associated with a position $\mathbf{x}_i \in \mathbb{R}^2$ and an initial position $\mathbf{x}_i^{\text{init}}$. In the ADMM formulation, $\mathbf{z}_i$ denotes the auxiliary variable representing the legalized position, and $\mathbf{u}_i$ is the scaled dual variable. The board region is denoted by B, and $\mathcal{F} \subseteq C$ represents the set of fixed components. Each component c_i is modeled as a union of convex sub-shapes $\mathcal{S}_i = \{s_i^1, \ldots, s_i^{m_i}\}$ to accurately capture irregular geometries.

2.3 Geometric Modeling and Collision Detection

We adopt the Separating Axis Theorem (SAT) for collision detection between convex polygons. Given two convex polygons P_A and P_B, they are non-overlapping if and only if there exists an axis $\hat{\mathbf{n}}$ such that their projections onto $\hat{\mathbf{n}}$ are disjoint. In PCB design, clearance constraints require maintaining a minimum distance between geometric objects, including component bodies, pads, and nets. To incorporate clearance into the SAT framework, we expand the projection intervals of each shape by the required clearance margin, so that the overlap computed on the expanded intervals reflects both collision and clearance violations in a unified manner. This allows collision detection and clearance enforcement to be handled consistently within the same geometric framework, avoiding separate constraint-handling mechanisms.

2.4 Minimum Translation Vector (MTV)

The minimum translation vector (MTV) is used to resolve collisions between overlapping convex shapes and represents the smallest displacement required to separate two intersecting convex polygons. Within the SAT framework, if no separating axis exists, the shapes overlap. For each candidate axis $\hat{\mathbf{n}}$, we compute the projected intervals of the two polygons and evaluate the overlap along that axis. The MTV is determined by selecting the axis with the minimum overlap:

$$\text{MTV} = \delta^* \hat{\mathbf{n}}^*, \tag{3}$$

where δ^* is the minimum overlap and $\hat{\mathbf{n}}^*$ is the corresponding axis, as illustrated in Fig. 3. The MTV provides both the direction and magnitude of the displacement required to resolve the collision. Compared with discrete constraint formulations, it provides a continuous approximation

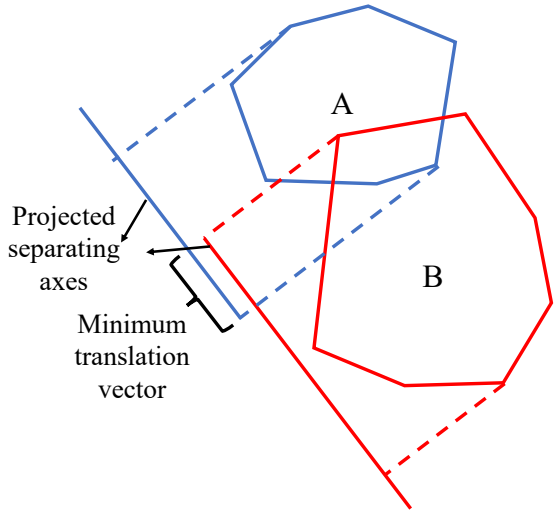


Figure 3: Illustration of collision detection and resolution using the Separating Axis Theorem (SAT) and minimum translation vector (MTV). Two overlapping convex polygons are projected onto candidate separating axes. Since no separating axis exists, the overlap is computed along each axis. The axis with the minimum overlap determines the MTV, which provides the smallest displacement required to separate the two shapes.

of geometric violations, making it naturally compatible with the ADMM framework, where violations are progressively reduced through iterative updates rather than resolved in a combinatorial manner.

2.5 Problem Formulation

In modern PCB designs, components have irregular geometries and must satisfy boundary, non-overlapping, and clearance constraints. The PCB legalization problem can be formally defined as follows.

PCB Legalization Problem. Given an initial placement $\{\mathbf{x}_i^{\text{init}}\}$, the goal is to determine a set of positions $\{\mathbf{x}_i\}$ that minimizes total displacement while reducing geometric violations, including overlap, boundary, and clearance violations.

3 PROPOSED ALGORITHM

This section presents the proposed consensus-optimization legalization framework, as illustrated in Fig. 4.

Given an overlapped initial placement together with board geometry and clearance rules, the framework proceeds in three phases: a geometric pre-separation stage that removes severe initial collisions to provide a stable starting point, a main ADMM optimization loop that decouples displacement minimization from geometric feasibility via variable splitting, and a final pair-based refinement stage that eliminates any remaining local conflicts.

The main ADMM loop performs three coordinated updates per iteration. The **x**-update minimizes displacement while enforcing board-boundary and fixed-component constraints; the **z**-update restores geometric feasibility through sequential Gauss–Seidel-based collision resolution, where violations are evaluated and corrective displacements are computed using SAT-based MTV; and the dual update reduces the discrepancy between the placement and auxiliary variables, driving the solution toward consensus. The loop terminates once the primal residual falls below a tolerance, and a final pair-based refinement stage resolves any remaining geometric or clearance conflicts using symmetric displacement, where both components share the adjustment to avoid bias and preserve the global placement structure.

3.1 ADMM Variable Splitting

To decouple displacement minimization from geometric feasibility, we introduce an auxiliary variable **z** and rewrite the legalization problem in

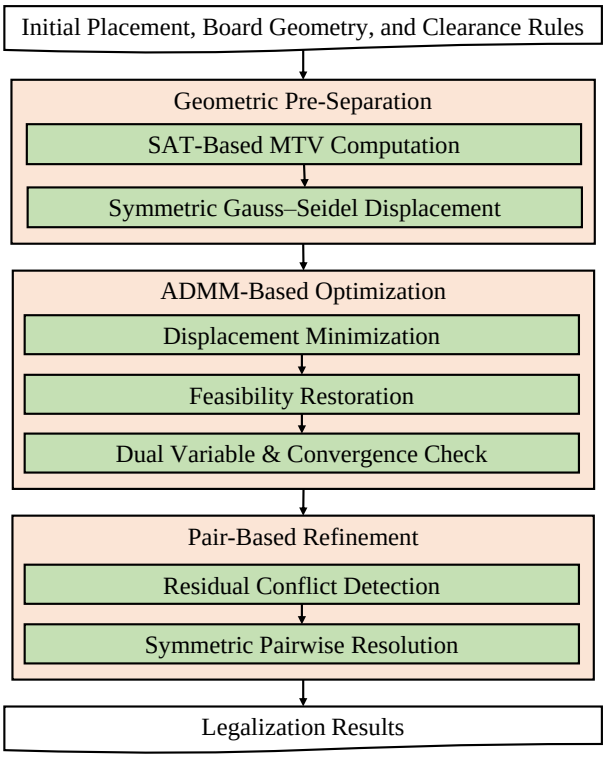


Figure 4: Our algorithm flow.

the consensus ADMM form:

$$\begin{aligned} \min_{\mathbf{x},\mathbf{z}} \quad & f(\mathbf{x}) + g(\mathbf{z}) \\ \text{s.t.} \quad & \mathbf{x} = \mathbf{z}, \end{aligned} \tag{4}$$

where

$$f(\mathbf{x}) = \sum_{c_i \in C} \|\mathbf{x}_i - \mathbf{x}_i^{\text{init}}\|_2^2 + \delta_{B\cap\mathcal{F}}(\mathbf{x}), \tag{5}$$

$$g(\mathbf{z}) = \delta_{\text{feasible}}(\mathbf{z}). \tag{6}$$

Here, $\delta_{B\cap\mathcal{F}}(\mathbf{x})$ is the indicator function that enforces board-boundary and fixed-component constraints, while $\delta_{\text{feasible}}(\mathbf{z})$ denotes the indicator function of the geometrically feasible region, which explicitly enforces non-overlap, clearance, and boundary constraints through SAT-based geometric modeling. Using the scaled dual variable $\mathbf{u}$, the augmented Lagrangian is written as

$$L_\rho(\mathbf{x},\mathbf{z},\mathbf{u}) = f(\mathbf{x}) + g(\mathbf{z}) + \frac{\rho}{2}\|\mathbf{x}-\mathbf{z}+\mathbf{u}\|_2^2 - \frac{\rho}{2}\|\mathbf{u}\|_2^2, \tag{7}$$

which explicitly decouples displacement minimization and geometric feasibility into separate subproblems.

3.2 Geometric Pre-Separation

Before entering the ADMM iterations, we perform a geometric pre-separation stage to remove large initial overlaps, serving as a warm start that improves the stability of the ADMM loop. Instead of dynamically updating all overlapping neighbors, this stage first identifies a fixed set of initially overlapping component pairs and processes only these pairs, which prevents the cascade effect where newly created overlaps introduce additional forces and destabilize the system. For each pair, SAT is used to compute an MTV, and the displacement is applied symmetrically to both components in an alternating Gauss–Seidel manner, so that each displacement is immediately reflected in subsequent computations. Compared with Jacobi-style simultaneous updates, this avoids force cancellation and improves stability in dense layouts, and a small overshoot factor κ is applied to each displacement to further reduce oscillation. Unlike the subsequent ADMM $\mathbf{z}$-update—which performs sequential feasibility restoration over the current placement and may include an additional clearance-enforcement pass after all body-level overlaps are removed—the pre-separation phase resolves only large body-level overlaps and ignores clearance constraints.

3.3 ADMM-Based Optimization

At each iteration, three updates are performed in sequence: (1) an $\mathbf{x}$-update that minimizes displacement while respecting board and fixed-component constraints, (2) a $\mathbf{z}$-update that restores geometric feasibility through sequential collision resolution, and (3) a dual update that drives $\mathbf{x}$ and $\mathbf{z}$ toward consensus. The loop terminates when the primal residual is sufficiently small or when no further improvement is observed, and any remaining conflicts are handled in the final refinement stage.

3.3.1 $\mathbf{x}$*-Update (Quadratic Program).* Given $\mathbf{z}^{(k)}$ and $\mathbf{u}^{(k)}$, the $\mathbf{x}$-subproblem is

$$\mathbf{x}^{(k+1)} = \arg\min_{\mathbf{x}} \sum_{c_i \in C} \left(\|\mathbf{x}_i - \mathbf{x}_i^{\text{init}}\|_2^2 + \frac{\rho}{2}\|\mathbf{x}_i - (\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)})\|_2^2 \right), \tag{8}$$

subject to board-boundary and fixed-component constraints. This subproblem is separable across components; for each movable component, the unconstrained optimum is

$$\mathbf{x}_i^{(k+1)} = \frac{\mathbf{x}_i^{\text{init}} + \rho(\mathbf{z}_i^{(k)} - \mathbf{u}_i^{(k)})}{1+\rho}. \tag{9}$$

In practice, board-boundary constraints are incorporated as bound constraints, and all variables are solved together using a warm-started QP solver. After the $\mathbf{x}$-update, rigidly bound heatsink components are synchronized to preserve their relative offsets.

3.3.2 $\mathbf{z}$*-Update (Collision Resolution).* Given $\mathbf{x}^{(k+1)}$ and $\mathbf{u}^{(k)}$, the $\mathbf{z}$-subproblem seeks a geometrically feasible placement close to the ADMM anchor $\mathbf{t}_i^{(k)} = \mathbf{x}_i^{(k+1)} + \mathbf{u}_i^{(k)}$. The anchor is used to initialize $\mathbf{z}$, and the subproblem is approximately solved via sequential Gauss–Seidel-based feasibility restoration in two stages. First, body-level geometric overlaps are resolved using SAT-based collision detection and MTVs. After all geometric overlaps are eliminated, clearance constraints are enforced using the same SAT-based MTV framework with expanded projection intervals; this staged strategy improves robustness, as directly enforcing clearance in highly overlapped configurations may lead to instability. All updates are applied sequentially, so that each displacement immediately affects subsequent collision checks within the same sweep. This Gauss–Seidel scheme avoids force cancellation and leverages the local nature of geometric conflicts in PCB layouts: since most violations occur between nearby components, the implicit ordering breaks symmetry and progressively propagates corrections across the layout, which is particularly important in dense placements.

3.3.3 SAT-Based Minimum Translation Vector. For each candidate pair (c_i, c_j) passing the AABB pre-filter, we examine all sub-shape pairs $(s_i^a, s_j^b) \in \mathcal{S}_i \times \mathcal{S}_j$ and compute the corresponding MTV using SAT. Clearance constraints are seamlessly integrated into this process: instead of treating clearance separately, we expand the projection intervals of geometric primitives by the required clearance margins. Specifically, clearance is enforced in a net-aware and geometry-aware manner: (1) body-level interactions are handled without additional clearance margins, (2) pad-level interactions incorporate spacing requirements depending on net relationships, and (3) cross-layer interactions are primarily considered for through-hole (TH) pads.

Given two convex polygons A and B with clearance margin m, all edge normals of A and B are treated as candidate separating axes, and each projected interval is expanded by the required clearance margin. If a separating axis exists, the two shapes are non-overlapping; otherwise, the axis with the minimum overlap determines the MTV,

$$\text{MTV} = \delta^* \hat{\mathbf{n}}^*, \tag{10}$$

so that both geometric overlap and clearance violations are resolved within a unified SAT-based framework. For each conflicting pair, the corresponding displacement is applied immediately to the current placement, so that subsequent pair evaluations observe the updated positions within the

same sweep. In the body-overlap restoration stage, the update can be written abstractly as

$$(\mathbf{z}_i, \mathbf{z}_j) \leftarrow \mathrm{ResolvePair}\big((\mathbf{z}_i, \mathbf{z}_j), \mathrm{MTV}(c_i, c_j)\big), \tag{11}$$

where the pairwise correction is applied before processing the next conflicting pair. After all body-level overlaps are removed, additional sweeps enforce clearance constraints using the same framework.

3.3.4 Dual Update and Convergence Criteria. After the $\mathbf{x}$-update and $\mathbf{z}$-update, the scaled dual variable is updated by

$$\mathbf{u}_i^{(k+1)} = \mathbf{u}_i^{(k)} + \left(\mathbf{x}_i^{(k+1)} - \mathbf{z}_i^{(k+1)}\right). \tag{12}$$

To monitor convergence, we compute the primal and dual residuals:

$$r_p^{(k+1)} = \sqrt{\frac{1}{n} \sum_{c_i \in C} \|\mathbf{x}_i^{(k+1)} - \mathbf{z}_i^{(k+1)}\|_2^2}, \tag{13}$$

$$r_d^{(k+1)} = \rho \sqrt{\frac{1}{n} \sum_{c_i \in C} \|\mathbf{z}_i^{(k+1)} - \mathbf{z}_i^{(k)}\|_2^2}. \tag{14}$$

The ADMM loop terminates when the primal residual falls below the tolerance or when no further improvement is observed.

3.4 Pair-Based Refinement

Although the ADMM loop resolves most conflicts, a few residual geometric or clearance violations may remain in tightly packed local regions. To eliminate these, each conflicting pair is resolved by symmetrically splitting the MTV between the two components:

$$\mathbf{z}_i \leftarrow \mathbf{z}_i + \frac{\kappa}{2}\mathrm{MTV}(c_i, c_j), \qquad \mathbf{z}_j \leftarrow \mathbf{z}_j - \frac{\kappa}{2}\mathrm{MTV}(c_i, c_j). \tag{15}$$

The updates are applied immediately in a Gauss–Seidel manner, which avoids force cancellation and is particularly effective for the final set of tightly coupled conflicts. This stage handles both geometric overlaps and clearance violations when enabled.

3.5 Heatsink Rigid Binding

Some components are rigidly attached to the same heatsink and cannot move independently. For each bound component c_i, its position is defined by a fixed offset $\boldsymbol{\delta}_i$ from a master component c_{m_i}:

$$\mathbf{x}_i = \mathbf{x}_{m_i} + \boldsymbol{\delta}_i, \qquad \mathbf{z}_i = \mathbf{z}_{m_i} + \boldsymbol{\delta}_i. \tag{16}$$

After each $\mathbf{x}$-update and $\mathbf{z}$-update, bound components are re-synchronized with their master component to maintain the prescribed offsets. During the dual update, they are skipped since their positions are fully determined by the master component.

4 EXPERIMENTAL RESULTS

We implemented the proposed consensus-optimization legalizer and the baseline legalizer of Ou et al. [13] within a unified PCB placement framework using C++ with Python bindings. The initial placements are generated by the clearance-constrained PCB global placer proposed in [14], ensuring that both legalizers start from identical placement configurations. All experiments were conducted on a Linux workstation running Ubuntu 22.04 with an AMD EPYC 7702 64-Core Processor (2×64 cores, 3.35 GHz) and 188 GB of memory. The MIQP formulation of Ou et al. was solved using Gurobi Optimizer 11.0.3 [15], while the QP subproblems in our ADMM framework were solved using OSQP 1.0 [16].

Table 2: Benchmark statistics and layout characteristics

Benchmark	#C	#Pads	#TH Pads	Top Util. (%)	Bot Util. (%)	Irregular Board?
PCB1	195/11	464	99	68.1	19.4	Yes
PCB2	53/5	131	33	83.7	29.1	Yes
PCB3	83/4	206	31	67.0	32.4	Yes
PCB4	84/5	199	42	62.3	20.3	Yes
PCB5	209/3	594	55	36.8	40.5	No
PCB6	201/5	547	27	43.0	20.1	No
ice40	44/0	323	12	30.1	26.5	No
Hor	290/2	436	234	35.4	23.2	Yes
mod	452/0	758	313	41.9	1.2	No

We evaluated both legalizers on six industrial PCB benchmarks derived from real power-adapter and DC–DC converter designs. In addition, we include three open-source benchmarks from PCBench [17], namely ice40 (ice40_stm32_fsmc_test_pcb_fsmc_ice40), Hor (Horticulture Light EQ 3-Band 3-channel-led-driver-rev0.4), and mod (mod12003_mod12003). As summarized in Table 2, these benchmarks cover a wide range of utilization levels, pad complexities, and board geometries, including irregular board outlines and high-density placements. Here, #C, #Pads, and #TH Pads denote the numbers of components, total pads, and through-hole pads, respectively, while Top Util. and Bot Util. represent the area utilization on the top and bottom layers. The component count ranges from 44 to 452, making them representative medium-scale PCB designs.

To ensure a fair comparison, we align the experimental settings of the baseline with those reported by Ou et al. [13], including benchmark configurations, placement initialization, and design rule specifications, and tune key solver parameters for best performance under our framework. For the ADMM-based legalizer, we used $\rho = 80$, $\epsilon = 0.001$, and a maximum of 1000 iterations. Both methods are applied to the same global placement results and executed under comparable runtime conditions, so the comparison focuses solely on the effectiveness of the legalization stage.

4.1 Comparison with the Baseline of Ou et al.

Table 3 compares the proposed consensus-optimization legalizer with the approach of Ou et al. [13] in terms of overlap violations (Ov.), clearance violations (Cl.), their reduction percentages, displacement (Disp.), wirelength change (WL), and runtime (RT). The normalized values use Ou et al. as the baseline. We note that when the baseline reports zero displacement and no reduction, its solver failed to find a feasible incumbent within the runtime limit and returned the initial placement unchanged, rather than certifying it as already legal. The MIQP-based method of Ou et al. [13] represents the only prior optimization-based PCB legalization framework capable of handling irregular geometries and arbitrary orientations, and therefore serves as the closest available baseline.

Overall, the proposed framework significantly outperforms the baseline in reducing DRVs, achieving an average overlap reduction of 55.72% versus 18.92% and a clearance reduction of 21.61% versus 4.63%; the baseline even degrades clearance in several benchmarks, as indicated by negative reduction values. The gap widens with problem complexity: on high-utilization or highly irregular benchmarks, the baseline stagnates early, whereas the proposed framework keeps improving across iterations, indicating better scalability. This is clearest on the hardest, highly infeasible cases: for PCB1–PCB3 the baseline produces essentially no improvement, while the proposed framework reduces overlap across all benchmarks, up to 83.28% on PCB6.

This improvement comes at the cost of larger displacement and moderate wirelength increase: the proposed framework incurs an average displacement of 125.51 versus 32.96 for the baseline and a 4.39% wirelength increase versus 1.27%. The proposed framework nonetheless achieves lower average runtime, because the baseline often spends substantial time exploring combinatorial search spaces without producing feasible updates, whereas the consensus-optimization framework avoids combinatorial branching.

4.2 Impact of Clearance-Aware Modeling

To analyze the effect of clearance constraints, Table 4 compares the proposed framework with and without explicit clearance enforcement, with normalized values relative to the variant without clearance. Without clearance-aware modeling, the framework achieves an average overlap reduction of 51.73% but a clearance reduction of −8.40%, showing that overlap removal alone may even worsen clearance quality. After incorporating clearance constraints, the overlap reduction slightly improves to 55.72%, while the clearance reduction significantly increases to 21.61%, with large improvements on PCB6 and PCB5 (67.82% and 47.95%, respectively).

Table 3: Comparison of Ou et al. [13] and the proposed consensus-optimization framework (Ours) with overlap/clearance reduction percentages

Benchmark	Initial		Ou et al. [13]							Ours						
	Ov.	Cl.	Ov.	Ov. Red.(%)	Cl.	Cl. Red.(%)	Disp. (mm)	WL(%)	RT(s)	Ov.	Ov. Red.(%)	Cl.	Cl. Red.(%)	Disp. (mm)	WL(%)	RT(s)
PCB1	368	115	368	0.00	115	0.00	0.00	0.00	57.96	167	54.62	125	-8.70	204.19	6.38	76.73
PCB2	84	41	84	0.00	41	0.00	0.00	0.00	6.97	49	41.67	54	-31.71	49.44	3.68	2.66
PCB3	180	103	180	0.00	103	0.00	0.00	0.00	7.33	73	59.44	78	24.27	96.05	9.65	7.82
PCB4	126	36	123	2.38	42	-16.67	9.69	0.34	1.51	39	69.05	35	2.78	49.91	4.47	14.25
PCB5	272	73	184	32.35	64	12.33	33.69	0.94	4.49	94	65.44	38	47.95	118.42	1.99	42.37
PCB6	299	174	195	34.78	94	45.98	106.41	4.84	7.95	50	83.28	56	67.82	121.25	7.03	3.81
ice40	75	0	68	9.33	0	0.00	4.00	-0.02	2.93	72	4.00	0	0.00	5.18	-1.68	0.17
Hor	640	0	55	91.41	0	0.00	136.82	3.29	209.65	189	70.47	18	N/A	135.12	2.90	0.93
mod	981	44	981	0.00	44	0.00	0.00	0.00	228.30	456	53.52	13	70.45	350.07	5.09	4.92
Average	336.11	65.11	248.67	18.92	55.89	4.63	32.96	1.27	58.34	132.11	55.72	46.33	21.61	125.51	4.39	17.07
Norm.	–	–	1.00	–	1.00	–	1.00	1.00	1.00	0.53	–	0.83	–	3.81	3.46	0.29

Table 4: Comparison of the proposed legalization results with and without clearance-aware modeling

Benchmark	Ours (w/o Clearance)							Ours (with Clearance)						
	Ov.	Ov. Red.(%)	Cl.	Cl. Red.(%)	Disp. (mm)	WL(%)	RT(s)	Ov.	Ov. Red.(%)	Cl.	Cl. Red.(%)	Disp. (mm)	WL(%)	RT(s)
PCB1	168	54.35	162	-40.87	184.33	5.97	7.54	167	54.62	125	-8.70	204.19	6.38	76.73
PCB2	41	51.19	52	-26.83	27.52	3.10	0.61	49	41.67	54	-31.71	49.44	3.68	2.66
PCB3	96	46.67	106	-2.91	77.73	6.96	2.74	73	59.44	78	24.27	96.05	9.65	7.82
PCB4	39	69.05	54	-50.00	40.72	3.68	3.02	39	69.05	35	2.78	49.91	4.47	14.25
PCB5	97	64.34	59	19.18	101.03	2.26	2.03	94	65.44	38	47.95	118.42	1.99	42.37
PCB6	62	79.26	129	25.86	103.10	6.52	0.43	50	83.28	56	67.82	121.25	7.03	3.81
ice40	68	9.33	0	0.00	4.00	-0.02	2.93	72	4.00	0	0.00	5.18	-1.68	0.17
Hor	55	91.41	0	0.00	136.82	3.29	209.65	189	70.47	18	N/A	135.12	2.90	0.93
mod	981	0.00	44	0.00	0.00	0.00	228.30	456	53.52	13	70.45	350.07	5.09	4.92
Average	178.56	51.73	67.33	-8.40	75.03	3.53	50.81	132.11	55.72	46.33	21.61	125.51	4.39	17.07
Norm.	1.00	–	1.00	–	1.00	1.00	1.00	0.74	–	0.69	–	1.67	1.24	0.34

This phenomenon reflects the intrinsic difference between overlap and clearance constraints: while overlap removal separates component bodies, clearance requires sufficient spacing between finer features such as pads and nets, so an overlap-free placement may still violate clearance rules at the pad level. By incorporating clearance into the SAT-based MTV computation, the proposed method addresses these fine-grained constraints at both the component and pad levels. The inclusion of clearance increases displacement (from 75.03 to 125.51) but, interestingly, lowers runtime (from 50.81 to 17.07), likely due to improved convergence after eliminating conflicting geometric configurations. This confirms that clearance-aware modeling is essential for practical PCB legalization.

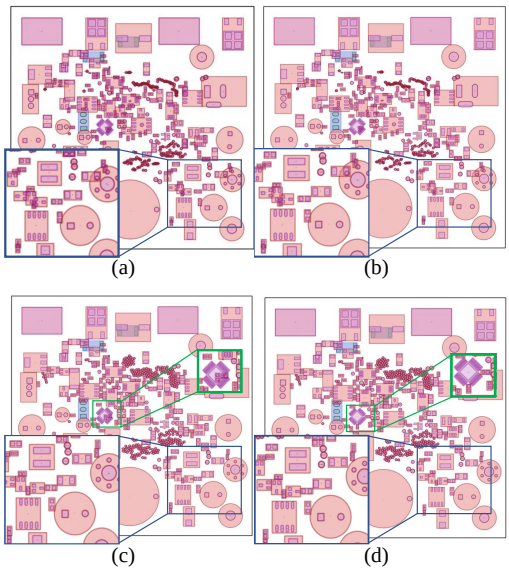


Figure 5: Visualization of legalization results on the *mod* benchmark. (a) Initial global placement with severe overlaps and clearance violations. (b) Legalization by Ou et al., which shows limited improvement in the placement. (c) The proposed legalization without clearance modeling, where overlaps are reduced in many cases but clearance violations remain. (d) The proposed legalization with clearance-aware modeling, which effectively resolves both overlap and clearance violations. The blue boxes highlight representative regions with severe overlap, while the green boxes indicate regions with significant clearance violations, where the differences among the methods are more clearly observed.

Fig. 5 shows a visual comparison on the *mod* benchmark: the baseline of Ou et al. leaves most overlaps unresolved (b); the proposed method without clearance modeling reduces overlaps but leaves clearance violations (c); and with clearance-aware modeling it resolves both, yielding a significantly cleaner, near-feasible placement (d), consistent with Table 4.

5 CONCLUSIONS

This paper presents a consensus-optimization framework for clearance-aware PCB legalization of heterogeneous and irregularly shaped components under complex geometric and design-rule constraints. Rather than treating geometric feasibility, clearance satisfaction, and displacement minimization as a single tightly coupled problem, the framework decomposes legalization into coordinated subproblems and drives them toward a globally consistent legal placement; ADMM serves as the underlying solver, while SAT-based minimum translation vectors enable accurate handling of exact irregular geometries and strict clearance requirements.

Experimental results on industrial PCB benchmarks demonstrate that the proposed method consistently achieves significantly fewer design rule violations than the MIQP-based baseline of Ou et al. [13] while maintaining competitive runtime, at the cost of increased displacement and wirelength. Unlike exact global optimization approaches, the framework progressively refines the placement through iterative optimization and geometry-aware conflict resolution: instead of requiring a globally feasible solution, it continuously reduces overlaps and clearance violations at each iteration, enabling meaningful improvements even in highly congested layouts where complete legalization may not be attainable within practical runtime budgets. This makes it a practical and scalable solution for modern PCB designs and a promising foundation for future integration with routing-aware and manufacturing-aware optimization techniques.

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